Microservices-Based E-Commerce Application

Master’s in Computer Engineering - Mobile Computing

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GitHub – E-Commerce

Project Overview

Transform a monolithic e-commerce app into a microservices system.

Goals:

- Modularity
- Scalability
- Maintainability

Features:

- Browse products
- Manage accounts
- Place orders
- PayPal integration
- Real Time Notifications

Architecture Concept

Independent services

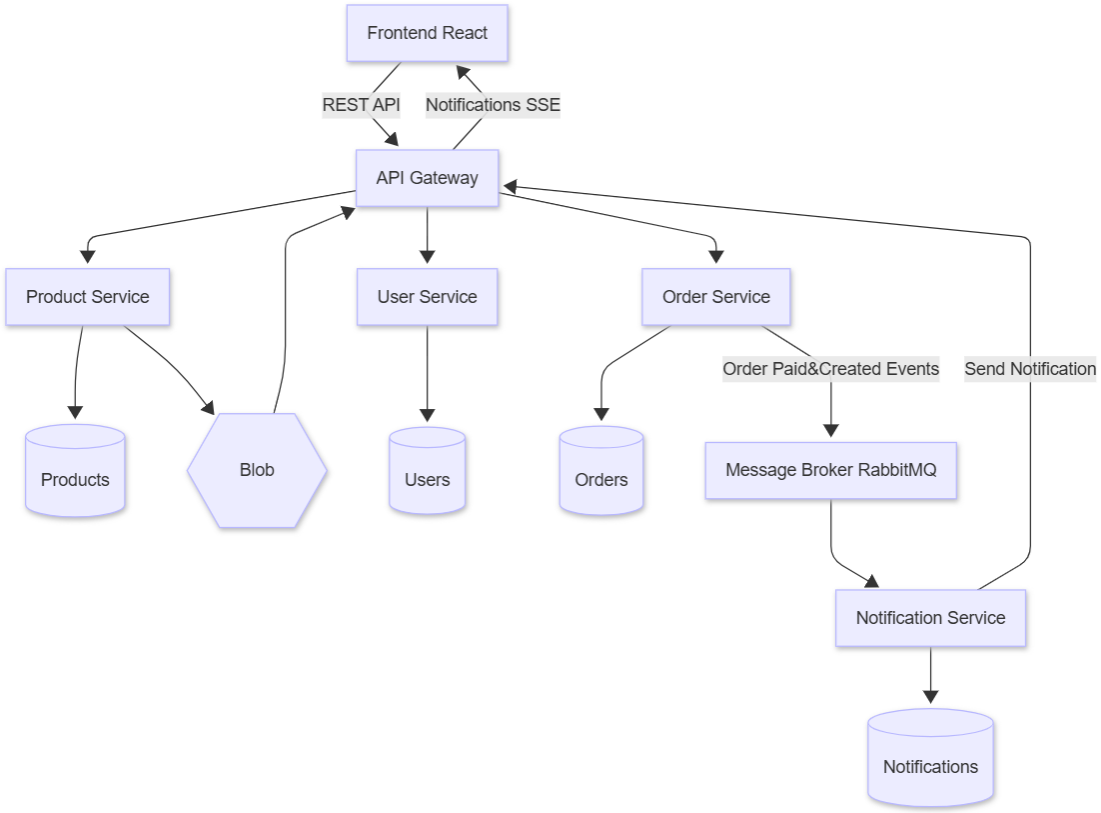
Communication:

- REST APIs
- Asynchronous messaging (Events)

Single entry point:

- API Gateway (nginx)

Architecture Overview



Microservices Breakdown

User Service: authentication & users

Product Service: catalog & images

Order Service: order processing

Notification Service: real-time updates

Frontend

Built with React

Enhancements:

- PayPal integration
- Notifications module
- Image retrieval via API Gateway

Backend Infrastructure

MongoDB (shared)

nginx API Gateway

Azurite (Blob storage emulator)

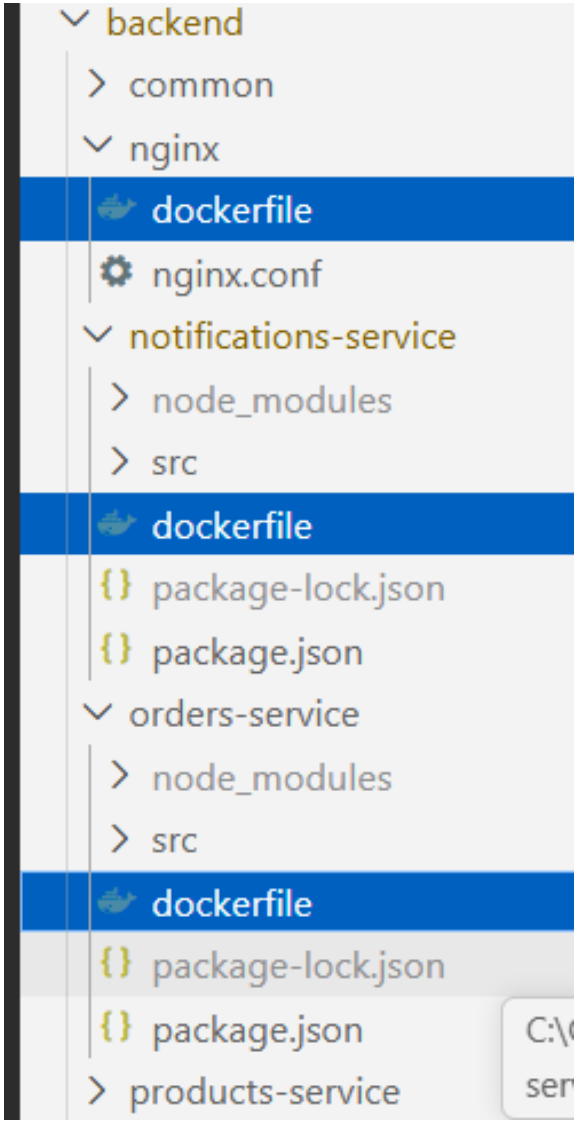
RabbitMQ (messaging)

Notifications System

Uses Server-Sent Events (SSE)

Flow:

- 1. Order created
- 2. Event sent to RabbitMQ
- 3. Notification service processes
- 4. Frontend receives update



Deployment with Docker

Each service runs in containers and has its own dockerfile.

Backend, shares common project reused by services.

Configs:

- Production
  - `docker compose -f docker.compose.yml up --build -d`
- Development
  - `docker compose -f docker.compose.yml -f docker.compose.dev.yml up --build`

System Orchestration

Containers:

- MongoDB
- Services
- RabbitMQ
- API Gateway
- Frontend

Challenges

Service startup timing

Reuse app common code without proper packing

Container networking

Blob storage access

SSE implementation

Possible Improvements

Add health checks

Use dedicated API Gateway

Use real cloud storage

Use npm packages to reduce dockerfile complexity.

Improve logging & monitoring

Conclusion

Microservices improve scalability and structure

Docker simplifies deployment

Practical experience with distributed systems

Final Note

A practical project focused on understanding microservices concepts and real-world challenges